

Novel natural killer cell-based therapies for hematologic and solid malignancies: latest updates from ASCO 2024.

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Natural killer (NK) cell-based therapies have made great progress in treating both hematological and solid tumors. Their unique mechanism of action does not rely on antigen presentation to recognize and eliminate tumor cells, making them a promising approach for cancer immunotherapy. In this review, we present a comprehensive summary of the latest clinical data of the novel NK cell-based therapies from the 2024 American Society of Clinical Oncology (ASCO) Annual Meeting, highlighting the potential of these advancements to revolutionize the treatment of hematologic malignancies and solid tumors.

NK cell receptors in anti-tumor and healthy tissue protection: Mechanisms and therapeutic advances.

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Natural Killer (NK) cells are integral to the innate immune system, renowned for their ability to target and eliminate cancer cells without the need for antigen presentation, sparing normal tissues. These cells are crucial in cancer immunosurveillance due to their diverse array of activating and inhibitory receptors that modulate their cytotoxic activity. However, the tumor microenvironment can suppress NK cell function through various mechanisms. Over recent decades, research has focused on overcoming these tumor escape mechanisms. Initially, efforts concentrated on enhancing T cell activity, leading to impressive results with immunotherapeutic approaches aimed at boosting T cell responses. Nevertheless, a substantial number of patients do not benefit from these treatments and continue to seek effective alternatives. In this context, NK cells present a promising avenue for developing new treatments, given their potent cytotoxic capabilities, safety profile, and activity against T cell-resistant tumors, such as those lacking HLA-I expression. Recent advancements in immunotherapy include strategies to restore and amplify NK cell activity through immune checkpoint inhibitors, cytokines, adoptive NK cell therapy, and CAR-NK cell technology. This review provides a comprehensive overview of NK cell receptors, the tumor escape mechanisms that hinder NK cell function, and the evolving field of NK cell-based cancer immunotherapy, highlighting ongoing efforts to develop more effective and targeted cancer treatment strategies.

CD3-negative natural killer cells express zeta TCR as part of a novel molecular complex.

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Natural killer (NK) cells are large granular lymphocytes capable of killing tumour cells in a non-MHC restricted manner. NK cells do not express cell-surface CD3, or any known target recognition structure analogous to the T cell antigen receptor (TCR) heterodimers (alpha beta or gamma delta). Consistent with their lack of expression of a CD3-TCR complex, NK cells do not require prior sensitization or antigen presentation by accessory cells to specifically recognize their tumour targets. Although NK cells do not express CD3-TCR, they do express CD2, the target of an alternative activation pathway which is functional in both T cells and NK cells. In T cells, this alternative activation pathway utilizes some component of the CD3-TCR complex as a transducer molecule that is required for mitogenesis. The fact

that NK cells are activated by this alternative pathway suggested that they might express a related subunit of the CD3-TCR complex capable of transducing the CD2-mediated signal. Here we show that human NK cells express the zeta-chain of the TCR complex in association with additional structures not included in CD3-TCR.

Natural killer cells and immunity against cancer.

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Extract: Under specific conditions, the immune response can eradicate tumors. The specific response against cancer requires the recognition of target molecules (antigens) by a subpopulation of white blood cells, the T lymphocytes. These antigens are exclusively or predominantly expressed by the tumor cells. Since many tumors fail to express specific antigens or express these antigens in a "wrong" way that impairs their recognition, cancer cells frequently escape from the specific immune response. Nonetheless, alternate effector cell populations have been described that are able to destroy tumor cells without any requirement for recognition of cancer-specific antigens. Among these effector populations, natural killer (NK) cells have an important role. Instead of looking for the presence of tumor cell-specific antigens, the NK cells sense the absence of molecules usually present at the surface of normal cells (the so-called human leukocyte antigen (HLA) class I molecules). Very often, these HLA class I molecules are not expressed, or are expressed but in an abnormal way by cancer cells. This "negative" mechanism of tumor cell recognition and destruction is called the "missing self hypothesis" since it postulates that the absence of molecules that identify the cell as "self" induces the destruction of these cells.

Natural killer cell-based strategies for immunotherapy of cancer.

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Natural killer (NK) cells are a family of lymphocytes with a natural ability to kill infected, harmed, or malignantly transformed cells. As these cells are part of the innate immunity, the cytotoxic mechanisms are activated upon recognizing specific patterns without prior antigen sensitization. This recognition is crucial for NK cell function in the maintenance of homeostasis and immunosurveillance. NK cells not only act directly toward malignant cells but also participate in the complex immune response by producing cytokines or cross-talk with other immune cells. Cancer may be seen as a break of all immune defenses when malignant cells escape the immunity and invade surrounding tissues creating a microenvironment supporting tumor progression. This process may be reverted by intervening immune response with immunotherapy, which may restore immune recognition. NK cells are important effector cells for immunotherapy. They may be used for adoptive cell transfer, genetically modified with chimeric antigen receptors, or triggered with appropriate antibodies and other antibody-fragment-based recombinant therapeutic proteins tailored specifically for NK cell engagement. NK cell receptors, responsible for target recognition and activation of cytotoxic response, could also be targeted in immunotherapy, for example, by various bi-, tri-, or multi-specific fusion proteins designed to bridge the gap between tumor markers present on target cells and activation receptors expressed on NK cells. However, this kind of immunoactive therapeutics may be developed only with a deep functional and structural knowledge of NK cell receptor: ligand interactions. This review describes the recent developments in the fascinating protein-engineering field of NK cell immunotherapeutics.

Roles of natural killer cells in immunity to cancer, and applications to immunotherapy.

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Great strides have been made in recent years towards understanding the roles of natural killer (NK) cells in immunity to tumours and viruses. NK cells are cytotoxic innate lymphoid cells that produce inflammatory cytokines and chemokines. By lysing transformed or infected cells, they limit tumour growth and viral infections. Whereas T cells recognize peptides presented by MHC molecules, NK cells display receptors that recognize stress-induced autologous proteins on cancer cells. At the same time, their functional activity is inhibited by MHC molecules displayed on such cells. The enormous potential of NK cells for immunotherapy for cancer is illustrated by their broad recognition of stressed cells regardless of neoantigen presentation, and enhanced activity against tumours that have lost expression of MHC class I owing to acquired resistance mechanisms. As a result, many efforts are under way to mobilize endogenous NK cells with therapeutics, or to provide populations of ex vivo-expanded NK cells as a cellular therapy, in some cases by equipping the NK cells with chimeric antigen receptors. Here we consider the key features that underlie why NK cells are emerging as important new additions to the cancer therapeutic arsenal.

Antigenic, functional, and molecular genetic studies of human natural killer cells and cytotoxic T lymphocytes not restricted by the major histocompatibility complex.

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Cytotoxicity not restricted by the major histocompatibility complex (MHC) is mediated by two distinct types of lymphocyte: natural killer (NK) cells and non-MHC-restricted cytotoxic T lymphocytes (CTL). These two types of cytotoxic lymphocytes can be distinguished by antigenic phenotype, function, and molecular genetic studies. In human peripheral blood, NK cells are identified by expression of the Leu-19 and/or CD16 cell surface antigens, and lack of CD3/T cell antigen receptor (Ti) complex expression (i.e., CD3⁻,Leu-19⁺). Peripheral blood non-MHC-restricted CTL express both CD3 and Leu-19 (i.e., CD3⁺, Leu-19⁺, referred to as Leu-19⁺ T cells). Both Leu-19⁺ T cells and NK cells lyse "NK-sensitive" hematopoietic tumor cell targets, such as K562, without deliberate immunization of the host. However, most "NK activity" in peripheral blood is mediated by NK cells, because they are usually more abundant and more efficient cytotoxic effectors than Leu-19⁺ T cells. The cytolytic activity of both NK cells and Leu-19⁺ T cells against hematopoietic targets was enhanced by recombinant interleukin 2 (rIL 2). NK cells, but not peripheral blood Leu-19⁺ T cells, were also capable of lysing solid tumor cell targets after short-term culture in rIL 2. Southern blot analysis of NK cells revealed that both the T cell antigen receptor beta-chain genes and the T cell-associated gamma genes were not rearranged, but were in germ-line configuration. These findings indicate that NK cells are distinct in lineage from T lymphocytes and do not use the T cell antigen receptor genes for target recognition.(ABSTRACT TRUNCATED AT 250 WORDS)

Pharmacological targeting of natural killer cells for cancer immunotherapy.

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Natural killer (NK) cells are innate lymphocytes that rapidly respond to cancer cells without prior sensitization or restriction to the cognate antigen in comparison with tumor antigen-specific T cells. Recent advances in understanding NK-cell biology have elucidated the molecular mechanisms underlying the differentiation and maturation of NK cells, in addition to the control of their effector functions by investigating the receptors and ligands involved in the recognition of cancer cells by NK cells. Such clarification of NK-cell recognition of cancer cells also revealed the mechanism by which cancer cells potentially evade NK-cell-dependent immune surveillance. Furthermore, the recent clinical results of T-cell-targeted cancer immunotherapy have increased the expectations for new immunotherapies by targeting NK cells. However, the potential use of NK cells in cancer immunotherapy is not fully understood. In this review, we discuss the current evidence and future potential of pharmacological targeting of NK cells in cancer immunotherapy.

Deprivation of human natural killer cells and antitumor immune response.

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Cell-based immunotherapy has been given increased attention as a treatment for cancer. Human natural killer (NK) cells are resident lymphocyte populations. They exhibit potent antitumor activity without human leukocyte antigen matching and without prior antigen exposure. They also are a promising tool for immunotherapy of solid and hematologic cancers. However, most cancer patients do not have enough NK cells to induce an effective antitumor immune response. This demonstrates a need for a source of NK cells that can supplement the endogenous cell population. In this study, we derived induced pluripotent stem cells (iPSCs) from peripheral blood T-lymphocytes using Sendai virus vectors. Generated iPSCs exhibited monoclonal T cell receptors (TCR) rearrangement in their genome, a hallmark of mature terminally differentiated T cells. These iPSCs were differentiated into NK cells using a two-stage coculture system: iPSCs into hematopoietic CD34+ cells with feeder cells M210-B4 (ATCC, USA) and CD34+ cells into mature NK cells with AFT024 cells (ATCC, USA). Our results showed that iPSC-derived NK cells expressed CD56, CD16, NKp 44 and NKp 46, possessed high cytotoxic activity and produced high level of interferon- γ . Based on our data, derivation of NK cells from induced pluripotent stem cells should be considered in the treatment of oncologic diseases. This would allow for the development of cell therapy for cancer using immunologically compatible NK cells derived from iPSCs. This may contribute to a more efficient treatment of oncologic diseases in addition to traditional cancer treatment.

Natural killer cells for pancreatic cancer immunotherapy: Role of nanoparticles.

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Advanced pancreatic cancer patients have a dismal prognosis despite advances in integrative therapy. The field of tumor immunology has witnessed significant advancements for cancer treatment. However, immunotherapy for pancreatic cancer is not very effective due to its highly complex tumor microenvironment (TME). Natural killer (NK) cells are lymphocytes that play an important role in the innate immune system. NK cells do not require antigen pre-sensitization, nor are they confined by the major histocompatibility complex (MHC). NK cells have the potential to eliminate cancer cells through CAR-dependent and CAR-independent pathways, demonstrating reduced levels of systemic toxicity in the process. The availability of several potential sources of NK cells is an additional benefit that contributes to meeting the therapeutic criteria. Adding nanotechnology to enhance the functions of effector NK cells

is also an appealing strategy. This article primarily discusses various approaches recently been utilized to enhance the NK functions for the treatment of pancreatic cancer. In addition, new advances in boosting NK cell therapeutic efficacy by nanoparticle mediation are presented, with a focus on pancreatic cancer.
